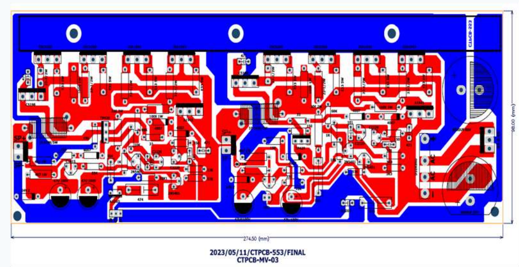
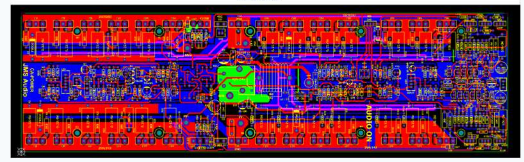
Project 1: High-Power Audio Amplifier Board

A high-fidelity, dual-channel (Stereo) high-power audio amplifier designed for premium sound reinforcement.

• Power Stage: Integrated complementary 2SC5200 & 2SA1943 transistors for high output.
• Power Subsystem: 6800µF / 80V filtering capacitors with heavy-duty bridge rectifier.
• Optimization: Linear component alignment for perfect heatsink attachment.

Ye dono alag alag Project hai to dono alag alag dikhana hai



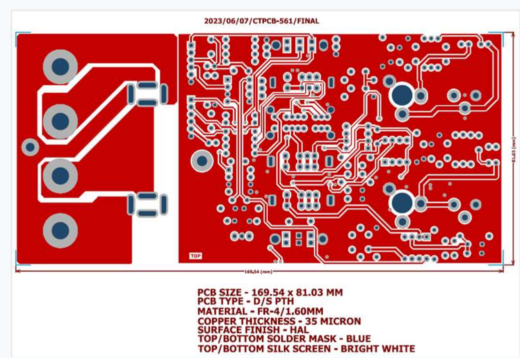


(Image 1 & 4 from uploaded files)

Project 2: Double-Sided High-Density Industrial Driver

A compact, high-density double-layer industrial-grade PCB optimized for automated manufacturing and high reliability.

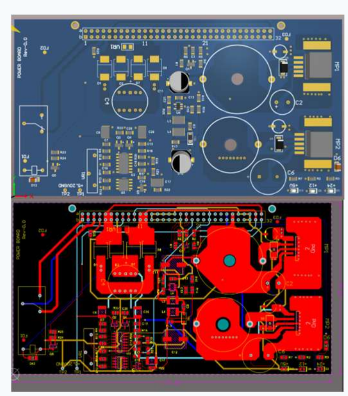
• Dimensions: 169.54 mm x 81.03 mm
• Stackup & Material: FR-4 substrate (1.60 mm) with 35 Micron (1 oz) copper.
• Finish & Aesthetics: HAL surface finish, Blue solder mask with sharp white silkscreen.



Project 3: Multi-Rail Power Supply & Distribution Board

A rugged industrial Power Board (Rev-0.0) engineered for structured power distribution across modular racks.

• Multi-Rail Output: Highly stable +5V, +12V, and +24V DC outputs with diagnostic LEDs.
• Design: Heavy-duty bottom layer copper pours to minimize voltage drop.
• Interface: 64-pin DIN/Eurocard-style edge connector for sub-rack integration.



Project 4: Aquai V1.0 – ESP32 Smart IoT Relay Controller

An Internet of Things (IoT) connected smart automation controller designed for handling high-voltage industrial loads.

• Core wireless SoC: Driven by powerful ESP32-WROOM module (Wi-Fi/Bluetooth).
• Actuators: 6 heavy-duty electromagnetic relays with flyback protective diodes.
• Connectivity: Onboard pluggable terminal blocks for quick sensor interfacing.